

AI, Machine Learning, and Text in Finance

Lecture 16 — AI/ML Foundations for Financial Text

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What Do We Mean by “Artificial Intelligence”?

THE BIGGER PICTURE

“AI” is coined by John McCarthy (1956): making machines perform tasks that *would require intelligence if performed by humans*. The boundary is socially constructed — once a machine does it reliably, we call it “mere computation” (the **AI effect**). The field chases a moving target.

Our working hierarchy for this book:

- **AI** — the broad enterprise of building systems that perform cognitive tasks.
- **Machine learning** — the dominant contemporary methodology.
- **Deep learning / LLMs** — particular implementations of that methodology.

Learning Objectives for This Lecture

By the end of this lecture you should be able to

- Place AI/ML/DL/LLM in a nested hierarchy.
- Contrast symbolic vs. statistical AI.
- Identify where AI creates value in finance.
- Categorise sources of financial text.
- Trace NLP from bag-of-words to transformers.
- State scaled dot-product attention.

From Expert Systems to Learning Machines

1956–1980s: Expert systems

- Hand-crafted “if-then” rules encoding domain knowledge.
- A 1980s credit system: hundreds of rules for “high-risk applicant”.
- **Strength:** transparent — every decision traces to a rule.
- **Weakness:** brittle — rules proliferate, fail on the unanticipated.

Late 1980s onward: induced from data

- Intelligence *learned* from data, not *encoded* by hand.
- Larger datasets + faster hardware made empirical evaluation feasible.
- By the mid-2000s data-driven methods overtook symbolic ones wherever labeled data existed: speech, vision, spam, and eventually NLP.

The shift in one sentence

The brittleness of rule-based systems motivated a move toward approaches in which knowledge is **induced from data rather than encoded by hand**.

Symbolic vs. Statistical AI

Symbolic AI

Represents the world through discrete, human-readable **symbols**; reasons by applying explicit **rules**. Knowledge encoded by experts *before* deployment; static as new data arrive.

Home: logic programming, semantic networks, knowledge graphs.

Statistical AI

Represents the world through numerical **parameters** estimated from data; predicts by evaluating a **learned function**. Knowledge implicit in parameters; *changes* on retraining.

Not mutually exclusive — the hybrid is common in finance

A statistical model flags an anomalous transaction; a *deterministic* compliance rule decides the regulatory report. We lean statistical (markets are data-rich) but return to symbolic methods for interpretability (explainability chapter) and regulatory constraints.

THE BIGGER PICTURE

ML = algorithms that improve at a task through *experience* (exposure to data). The central desideratum is **generalization**: perform well not just on seen data but on new inputs from the same distribution.

UNDER THE HOOD

Supervised learning. Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, learn $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ by minimizing

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(\mathbf{x}_i), y_i),$$

with per-sample loss ℓ (cross-entropy, squared error). Covers *classification* and *regression* — the two workhorses of quant finance.

Three Learning Paradigms

THE BIGGER PICTURE

The same generalization goal recurs under three supervisory regimes, each mapping to a different class of financial problem.

- **Supervised:** credit scoring, default prediction, factor estimation — learn from labeled (x_i, y_i) pairs.
- **Unsupervised:** clustering trades into market regimes — structure without labels.
- **Reinforcement:** maximise cumulative reward (trading, rebalancing) — learn from interaction.

Why the learning paradigm matters in finance

A supervised credit model needs *labeled historical defaults* and is constrained to the distribution of its training period. A model fit on a benign credit cycle can badly **underestimate tail risk** in a crisis — an instance of **distribution shift** with practical *and regulatory* consequences.

Practical implications threaded through the book:

- Generalization challenges pervade financial ML.
- They motivate careful cross-validation (purging look-ahead bias).
- Reference: *Advances in Financial Machine Learning* (Lopez de Prado, 2018).

Deep Learning: Representation at Scale

Deep learning = ML with neural networks deep enough that the model *learns intermediate representations automatically* — no hand-specified features.

UNDER THE HOOD

Universal approximation: hierarchical compositions of simple nonlinear functions represent arbitrarily complex mappings given enough width/depth. *Depth* confers a specific advantage:

- Lower layers: general, transferable features (edges, phonemes, collocations).
- Higher layers: task-specific combinations.

This makes **transfer learning** efficient: pre-train broadly, fine-tune on a narrow task with far fewer labeled examples.

Deep Learning: The Market Analogy

THE BIGGER PICTURE

Deep nets $\leftrightarrow$ **markets**. Both aggregate information hierarchically across interacting units / layers, and both are hard to interpret from the outside — motivating the explainability methods later in the book.

Why depth matters in finance

The same hierarchy that makes deep nets powerful — transferable low-level features composed into task-specific representations — is what makes a single pre-trained backbone reusable across many financial text tasks, the theme of the pre-training and transfer-learning slides that follow.

Large Language Models as a Special Case

UNDER THE HOOD

A **token** is the atomic unit of text; the **vocabulary** $\mathcal{V}$ is the finite set of such units. An LLM is a network with *billions* of parameters, trained on *hundreds of billions* of tokens, to assign probabilities to token sequences. Canonical objective — next-token prediction:

$$\mathcal{L}_{\text{LM}} = - \sum_{t=1}^T \log P_{\theta}(x_t \mid x_1, \dots, x_{t-1}).$$

What Makes LLMs Special

What makes LLMs *special*: scale + architecture + **emergent abilities** — capabilities appearing abruptly with scale (Wei et al., 2022):

- Multi-step reasoning, instruction following.
- **In-context learning:** adapt to a task from a few prompt examples, no parameter update.

Consequence for finance

A *single* LLM is a flexible substrate for earnings-call summarization, regulatory parsing, and strategy explanation — e.g. BloombergGPT (Wu et al., 2023).

Decision-Making Under Uncertainty

THE BIGGER PICTURE

Finance is fundamentally *decision-making under uncertainty*: form beliefs about future states, then act subject to constraints. AI contributes at *two* levels.

Quality of beliefs

Better forecasts; richer models of uncertainty.

Quality of decisions

Portfolio optimization, dynamic programming, RL.

Reflexivity — finance is not medical imaging

The decision procedure can *reshape* the market and *invalidate* the forecast model. AI systems in finance are **actors within the market**, not mere observers — a strategic dimension with no analogue in most prediction problems.

Example: Reflexivity in Algorithmic Trading

Deep RL market maker

A market-making agent quotes bid/ask at high frequency, earning the spread while managing inventory risk. A **deep reinforcement learning** agent (NN policy maximising cumulative reward) trains across millions of simulated episodes, learning to adapt to changing liquidity.

The feedback loop:

agent's actions $\longrightarrow$ order book $\longrightarrow$ signals the agent conditions on $\longrightarrow$...

Design lesson

The loop must be modeled *explicitly* in the training environment — otherwise the policy performs well in simulation but **destabilizes in live markets**.

Forecasting, Risk, and Alpha Generation

Core quant problems — all transformed by ML. The common enemy is the **signal-to-noise ratio**: returns near-unpredictable short-horizon; risk models extrapolate into rare tails; alpha erodes once discovered.

Problem	ML contribution	Care required
Return forecasting	High-dim feature spaces; no pre-specified interactions	Purge look-ahead bias; control multiple comparisons
Risk estimation	Richer distributions; regime switching; cross-asset learning	Tails rarely observed; stress periods break Gaussian VaR
Alpha / signals	Text-based signals add info absent from price & volume	Genuine alpha erodes fast once public

Where text comes in

Variance-covariance VaR assumes Gaussian returns and stable correlations — assumptions that fail *precisely under stress*. Language models parse narrative risk disclosures, turning qualitative uncertainty into quantitative signals.

Regulatory Compliance and Fraud Detection

Beyond investment management — AI in operational & compliance functions:

- AML, sanctions screening, KYC, real-time fraud detection.
- All: spot *rare, anomalous* patterns in high-volume streams — a fit for classifiers trained on labeled historical incidents.

Why language models add particular value here

Much relevant information is *textual*: suspicious activity reports, correspondence, filings, analyst opinions. An LLM can surface unusual phrasing, flag narrative-vs-quantitative discrepancies, or help interpret new guidance.

Regulation constrains design

EU AI Act and GDPR's "right to explanation" can make **opaque models inadmissible** for certain decisions *regardless of accuracy* — the recurring performance-vs-interpretability tension.

Structured vs. Unstructured Data in Finance

Structured data

Rows & columns; well-defined types; each row a known entity at a time.

Prices, accounting items, volumes. Served well by regression, factor models, time series *since the 1960s*.

Unstructured data

No pre-defined schema — text, audio, images, video. Needs a *pipeline* to turn raw content into statistical features.

Limits of structured data: backward-looking; capture *outcomes* (prices, earnings) not the reasoning/sentiment behind them; observed by everyone simultaneously $\Rightarrow$ limited informational edge.

THE BIGGER PICTURE

A large institution may see **tens of thousands of documents per day** — too voluminous to read systematically. This **processing bottleneck** is a persistent opportunity for ML methods that scale.

Sources of Financial Text

Source	Content & evidence
Corporate filings	SEC EDGAR: 10-K, 10-Q, 8-K. Narrative MD&A, Risk Factors, forward-looking statements. Loughran–McDonald (2011) : 10-K sentiment (finance lexicon) predicts post-filing returns.
News & wires	High-frequency public information. Tetlock (2007) : pessimistic WSJ “Abreast of the Market” language predicts downward pressure then mean-reversion.
Earnings calls	Live: scripted remarks <i>plus</i> unscripted Q&A. Tone, question types, hedging, even speech pace carry incremental information beyond reported numbers.

Signal hidden in text

EPS \$1.20 *beats* the \$1.15 consensus — looks positive. But Risk Factors grew **30% YoY**, three new risks concern a politically sensitive supply chain, and the CEO said “challenging” **eleven times** — a record. The headline reader misses the signal.

Why Text is Hard: Noise, Ambiguity, Context

- 1 **Ambiguous.** Meaning depends on author, time, context. “Uncertainty” in a Fed statement $\neq$ in a founder’s letter. “Positive outlook” is bullish in equity research but *cautious* in credit ratings. $\sim 3/4$ of Harvard-Inquirer “negative” words are neutral/positive in filings (Loughran–McDonald) — hence a domain lexicon.
- 2 **High-dimensional & sparse.** A 50k-word vocabulary $\Rightarrow$ a document-term matrix with far more dimensions than observations. Needs dimensionality reduction (topic models, embeddings, LM encoders).
- 3 **Strategically produced.** Disclosures written by lawyers/comms aware of liability and market reaction. Taking text at face value gets you **systematically misled**.

THE BIGGER PICTURE

Context is a first-class variable. Bullish vs. bearish, compliant vs. misleading, consistent vs. contradictory — all need a rich representation of entity, time, regime, and genre. This long-range contextual reasoning is exactly what **transformers** are designed to do.

Early NLP in Finance: BoW, Dictionaries, Topic Models

Bag-of-Words (BoW) + tf-idf

Document = vector of word counts; *all* order and grammar discarded. tf-idf up-weights rare informative words. Applied to filings since the 1990s for tone and readability (Kearney & Liu, 2014).

Dictionary sentiment

Map words to sentiment scores; net sentiment = (positive – negative) frequency. **Loughran–McDonald lexicon:** ~2,700 words in 6 categories (negative, positive, uncertainty, litigious, strong/weak modal) — still a benchmark.

Topic models (LSA / LDA)

Learn representations from *co-occurrence* — each document a mixture over latent themes (e.g. operational efficiency vs. regulatory risk).

Shared limitation

Shallow semantics: words are atomic, meanings context-*independent*, word order carries no information.

The Transformer Revolution (Vaswani et al., 2017)

THE BIGGER PICTURE

“Attention Is All You Need” (2017) triggered the largest transition in NLP history. **Self-attention** lets each position attend to *every* other position — resolving long-range dependencies (a pronoun to its antecedent; a Risk Factor to a financial-statement figure) that recurrent nets, with vanishing gradients over long contexts, cannot do reliably.

The Transformer Revolution: Scaled Dot-Product Attention

UNDER THE HOOD

For each position i , a weighted sum of value vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V},$$

with $\mathbf{Q}, \mathbf{K}$ query/key projections (dim d_k), $\mathbf{V}$ the value projection. The $\sqrt{d_k}$ scaling keeps dot products from pushing the softmax into a saturated, vanishing-gradient regime. Multiple **heads** attend to different aspects (syntax, coreference) in parallel.

Pre-trained transformers — BERT, GPT, RoBERTa, successors — combine self-attention with *unsupervised pre-training* on very large corpora.

The transfer-learning economics

Rich, context-sensitive representations $\Rightarrow$ fine-tune on downstream tasks with *relatively small* labeled datasets. Tasks that once needed **thousands** of labeled examples now need **hundreds** — lowering the barrier for financial applications where labels are expensive.

- Lower layers learn transferable structure; higher layers specialise.
- Same backbone serves sentiment, NER, similarity, summarization, Q&A.
- Full attention derivation & financial NLP implications: LLM-foundations chapter.

Why LLMs are Particularly Suited to Finance

- ➊ **Long-document reasoning.** Context windows of tens–hundreds of thousands of tokens process a whole 10-K or call transcript in one pass — no information loss from chunking.
- ➋ **In-context learning.** Supply 3 examples of suspicious patterns, evaluate a new transaction — *few-shot*, valuable where labels are scarce.
- ➌ **World knowledge.** Geopolitics, supply chains, governance norms let the model contextualise a disclosure. BloombergGPT shows finance pre-training helps, but general frontier models already do well.
- ➍ **Natural-language interface.** “Summarize the key risks in this 10-K and flag changes from last year” — no code, democratising text analysis.

LLMs are not a panacea

They **hallucinate** (confident but wrong); are **prompt-sensitive**; their reasoning is **opaque**; and they may be **poorly calibrated** on rare, high-stakes events. Deploy carefully — evaluation, monitoring, human oversight.

Roadmap: What the Book Covers

Part I — Foundations

Transformer architecture; training & fine-tuning (SFT, RLHF, LoRA); multi-agent systems; business valuation (DCF, comparables).

Part II — Applications

Credit risk; frontier applications; domain-specific LLMs; financial NLP / sentiment; portfolio & quant trading; RegTech & AML; explainability. *Read independently.*

Part III — Deploy

LLM limitations & evaluation; financial text summarization; privacy-preserving / local models.

Reading conventions

Context boxes (“the bigger picture”) give intuition; *Deep Dive* boxes (“under the hood”) give the mechanism — skippable without loss. Exercises tagged **[B]** / **[I]** / **[A]**. Companion Python notebooks throughout.

Lecture 16 Summary

Definitions & hierarchy: $AI \supset ML \supset DL \supset LLM$ · symbolic vs. statistical · expert systems → learning from data.

ML mechanics: supervised loss minimization · generalization · distribution shift · deep representation learning · next-token objective · emergence & in-context learning.

Finance applications: beliefs vs. decisions · reflexivity · forecasting / risk / alpha · AML, KYC, fraud · regulation constrains design.

Text: structured vs. unstructured · filings / news / calls · ambiguity, sparsity, strategic production · context as first-class variable.

BoW → LLMs: tf-idf, LM dictionary, LDA · self-attention + $\sqrt{d_k}$ · pre-training / transfer · four finance-suited properties · **not a panacea**.

Next

Lecture continues into the transformer architecture from the ground up (LLM-foundations) — attention, pre-training objectives, tokenization.

APPENDIX

Definitions, Derivations, and Secondary Material

Extra / optional material — beyond the core lecture.

The AI Effect and the Shifting Definition

The AI effect (ratchet)

Tasks once seen as hallmarks of intelligence — chess, diagnosis, translation — get reclassified as “mere computation” once machines do them reliably. So the *boundary* of AI is socially constructed as much as technically defined, and the field perpetually chases a moving target.

Our pragmatic convention for the book: AI = building systems that perform cognitive tasks; ML = the dominant methodology; DL and LLMs = particular implementations. Avoids endless definitional debate while staying precise enough to work.

Supervised learning

Input space $\mathcal{X}$, output space $\mathcal{Y}$, parameters $\theta \in \Theta$, per-sample loss $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$. Minimise $\mathcal{L}(\theta) = \frac{1}{n} \sum_i \ell(f_\theta(\mathbf{x}_i), y_i)$; goal is generalization to unseen inputs.

Large language model

Network with billions of parameters, trained on hundreds of billions of tokens, assigning probabilities to token sequences; objective $\mathcal{L}_{\text{LM}} = -\sum_t \log P_\theta(x_t \mid x_{<t})$.

Structured vs. unstructured

Structured = relational rows/columns, typed fields, known entity-at-time. Unstructured = no schema (text, audio, image, video); needs a feature pipeline.

Multi-Head Attention and the $\sqrt{d_k}$ Scaling

UNDER THE HOOD

Why scale by $\sqrt{d_k}$? If query/key components are $\sim (0, 1)$ i.i.d., $\text{Var}(\mathbf{q}^\top \mathbf{k}) = d_k$. Large logits push softmax into a saturated regime where gradients vanish. Dividing by $\sqrt{d_k}$ normalises variance to 1.

Multi-head: run h heads in parallel subspaces, then concatenate and project. One head may track *syntactic agreement* while another tracks *coreference*. In the standard single-head presentation $d_v = d_k$; multi-head implementations may differ.

Standard configuration (base Transformer)

$h = 8$ heads, $d_k = d_v = 64$, $d_{\text{model}} = 512$, $L = 6$ layers. Modern LLMs: $L = 32\text{--}96$, $d_{\text{model}} = 4,096\text{--}8,192$.

Frameworks that constrain financial AI design:

- **EU AI Act** — high-risk categories (credit scoring, insurance pricing); conformity assessment, human oversight, transparency.
- **GDPR right to explanation** — opaque models may be inadmissible for certain decisions regardless of accuracy.

Application areas (treated in depth later)

AML, sanctions screening, KYC, transaction fraud — rare-event detection over high-volume streams, much of it textual (suspicious activity reports, correspondence, filings). See the RegTech & AML chapter.